Program:

```
#include<stdio.h>
#include<malloc.h>
struct node
{
   int data;
   struct node *prev,*next;
};
struct node *head,*newnode,*temp,*tail,*before,*curr;
int total_elements = 0;

void delete(int data)
{
   int found = 0;
   before = NULL;
   curr = head;
   while(curr != NULL)    {
      if(curr->data == data)
      {
         found = 1;
         break;
      }
      before = curr;
      curr = curr->next;
   }
   if(found == 0)
   {
      printf("\n element not found \n");
```

```
   }
   else
   {
      if(curr == head)
      {
         head = curr->next;
         head->prev = NULL;
         free(curr);
      }
      else if(curr->next == NULL)       {
         before->next = NULL;
         tail = before;
         free(curr);
      } else {
         before->next = curr->next;
         curr->next->prev = before;
         free(curr);
      }
      total_elements--;
     printf("the element deleted: %d \n",data);
   }
}
void prepend(int data)
{
   newnode = (struct node*)malloc(sizeof(struct node));
   newnode->data = data;
   newnode->next = newnode->prev = NULL;
```

```
    if(head == NULL)    {
       head = newnode;
    } else  {
       newnode->next = head;
       head->prev = newnode;
       head = newnode;
    }
    total_elements++;
      printf("the element prepended: %d \n",data);
}

void append(int data)
{
    newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = data;
    newnode->next = newnode->prev = NULL;
    if(head == NULL)    {
       head = newnode;
       tail = newnode;
    }
```

```
    else
    {
       temp = head;
       while(temp->next != NULL) {
          temp = temp->next;
       }
       temp->next = newnode;
       newnode->prev = temp;
       tail = newnode;
    }
    total_elements++;
    printf("the element appended: %d \n",data);
}
void display()
{
    temp = head;
    printf("\n NULL <->");
    while(temp != NULL)    {
       printf("%d <->",temp->data);
       temp = temp->next;
    }
    printf("NULL \n");
}
void reverse() {
    temp = tail;
    printf("\n NULL <->");
    while(temp != NULL)    {
       printf("%d <->",temp->data);
       temp = temp->prev;
    }
    printf("NULL\n");
}
int main() {
    append(10);
    append(20);
    append(30);
```

```
void display()
{
   temp = head;
   printf("\n NULL <->");
   while(temp != NULL)    {
      printf("%d <->",temp->data);
      temp = temp->next;
   }
   printf("NULL \n");
}
void reverse() {
   temp = tail;
   printf("\n NULL <->");
   while(temp != NULL)    {
      printf("%d <->",temp->data);
      temp = temp->prev;
   }
   printf("NULL\n");
}
int main() {
   append(10);
   append(20);
   append(30);
   display();
   prepend(40);
   display();
   printf("\n total elements:%d",total_elements);
   printf("\n----\n");
   delete(30);
   display();
   printf("\n total elements:%d",total_elements);
         reverse();
   return 0;
}
```

Output:

```
the element appended: 10
the element appended: 20
the element appended: 30

 NULL <->10 <->20 <->30 <->NULL
the element prepended: 40

 NULL <->40 <->10 <->20 <->30 <->NULL

 total elements:4
  ----
the element deleted: 30

 NULL <->40 <->10 <->20 <->NULL

 total elements:3
 NULL <->20 <->10 <->40 <->NULL
```